CLASS: SY-B ENTC

PRN: 25UET076

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SLA-4:

Problem Statement 11: Intelligent Cooperative Multi-Agent System

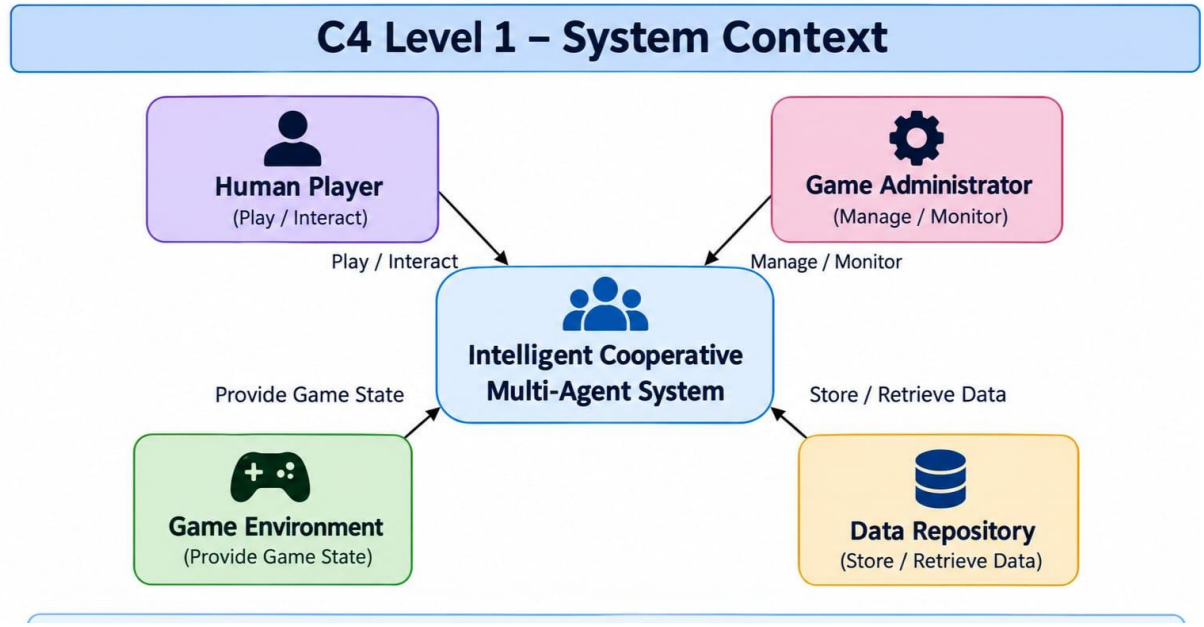
INTRODUCTION:

The Intelligent Cooperative Multi-Agent System is designed to improve teamwork and decision-making in games. In traditional games, multiple independent agents may make different or conflicting decisions because they have limited information and coordination. This system allows multiple intelligent agents to communicate with each other, share information, and coordinate their actions to achieve a common goal. Each agent observes the game environment, makes decisions, and cooperates with other agents. The system can also learn from previous game results and improve its strategies over time. The C4 architecture represents the system at four levels: System Context, Container, Component, and Code, making its structure and working easy to understand.

To solve this problem, the system allows agents to share information, communicate with each other and coordinate their actions. It also uses learning techniques to learn from previous experiences and improve future decisions. The system analyzes game conditions, selects suitable strategies and evaluates the results of its actions.

This project uses the C4 architecture model, which explains the system at four levels: System Context, Containers, Components and Code. These levels help us understand the overall system, its main parts and how its internal components work together.

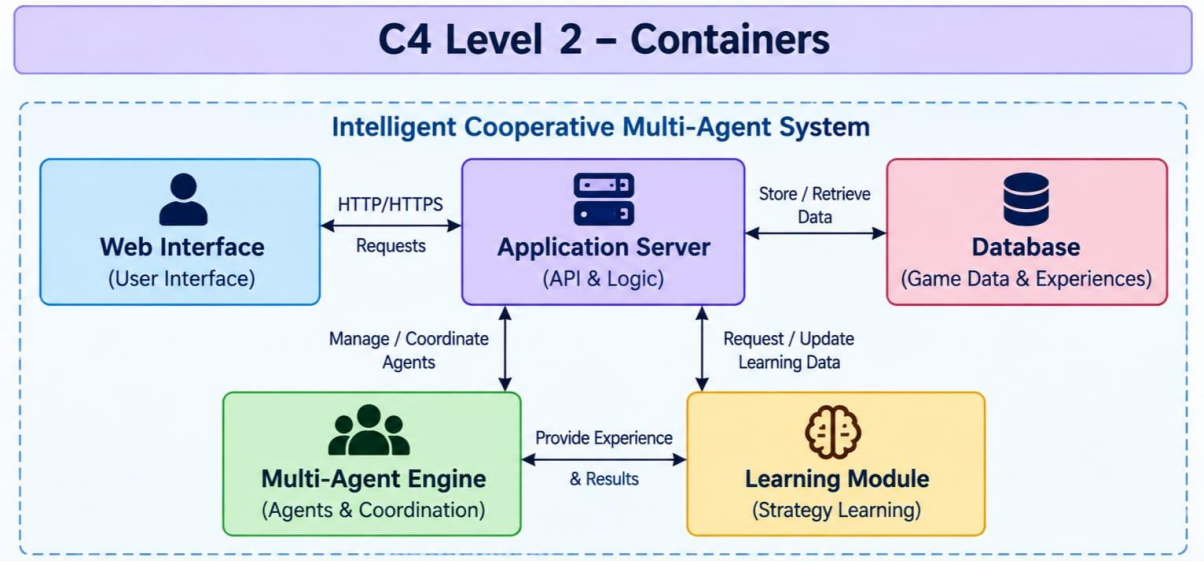
C4 LEVEL 1 :



This diagram shows the overall system and its interactions with external entities, such as the human player, game administrator, game environment and data repository. It provides a high-level overview of how users interact with the Intelligent Cooperative Multi-Agent System

The System Context diagram represents the complete system and its surroundings. It identifies the main users and external entities that interact with the system. It helps us understand who uses the system, how they interact with it, and what information they provide. This level gives a simple overview of the system without showing its internal technical details.

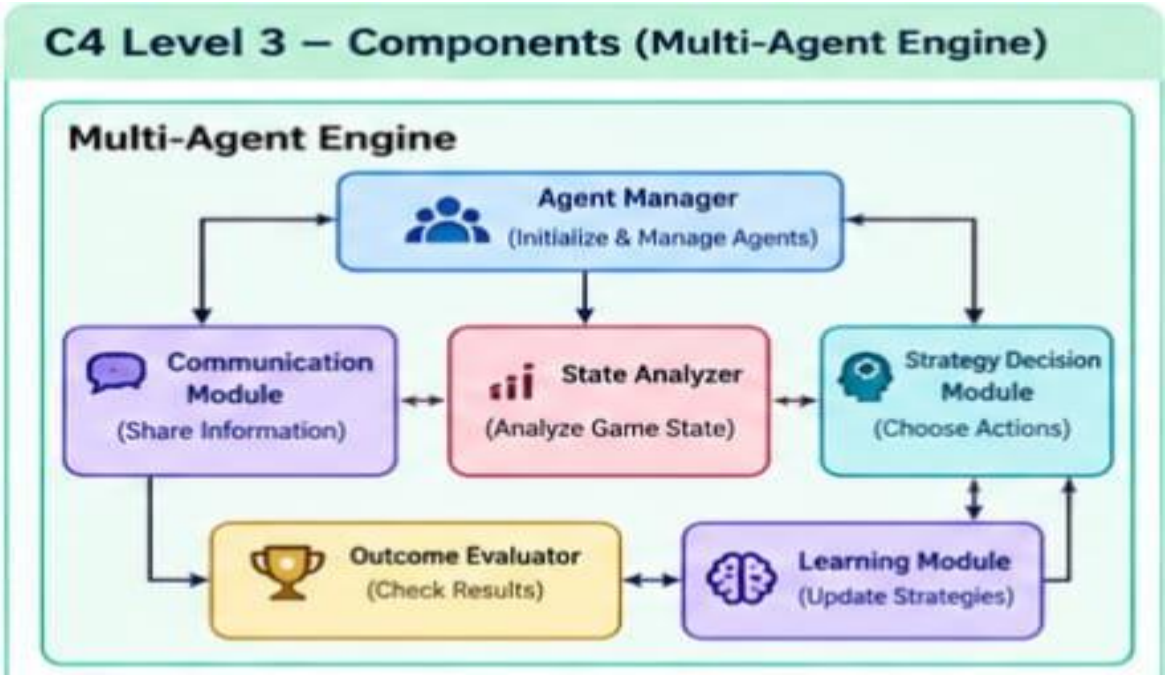
C4 LEVEL 2:



This diagram describes the major containers of the system, including the web interface, application server, multi-agent engine, learning module and database. The web interface allows users to interact with the system. The application server processes requests, while the multi-agent engine manages agent communication and coordination. The learning module improves strategies using previous game experiences, and the database stores game information and results.

The Container diagram explains how the system is divided into major functional parts. Each container performs a specific task and communicates with other containers. The Web Interface provides user interaction, the Application Server handles requests and logic, the Multi-Agent Engine manages agent coordination, the Learning Module improves strategies, and the Database stores game data and experiences. This level explains how the main parts work together.

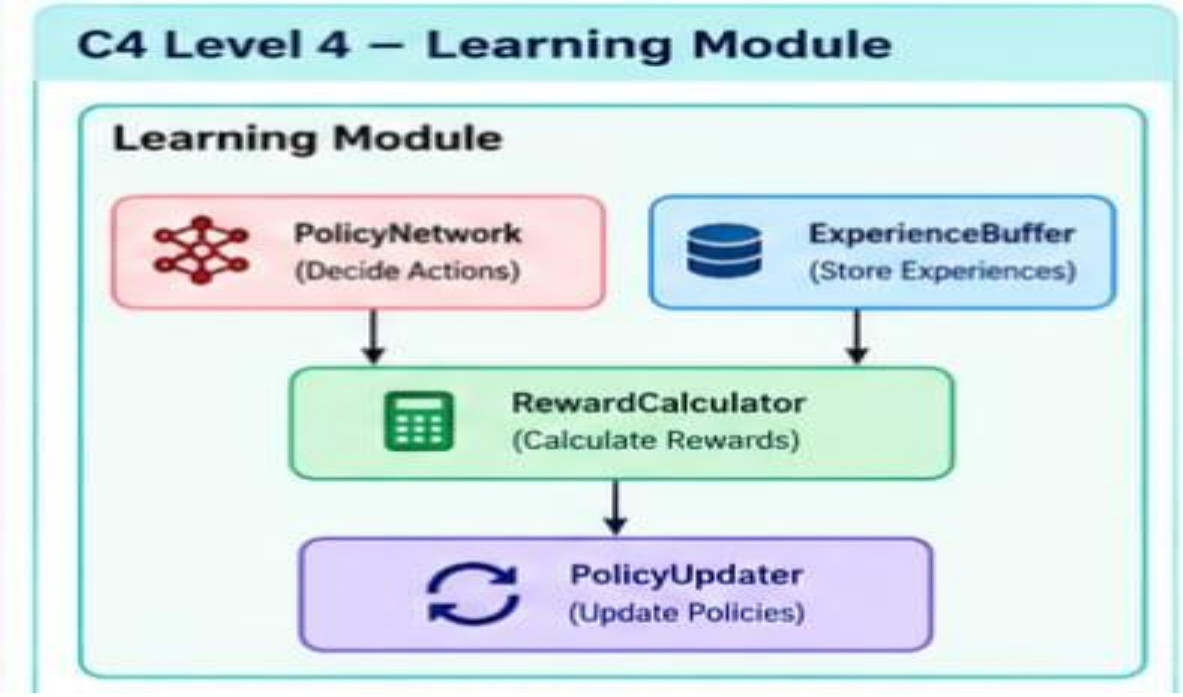
C4 LEVEL 3:



This diagram illustrates the internal components of the Multi-Agent Engine, such as the Agent Manager, Communication Module, State Analyzer, Strategy Decision Module, Outcome Evaluator and Learning Module. These components work together to analyze game conditions, coordinate agent actions and adapt strategies based on game outcomes

The Component diagram provides a detailed view of the internal structure of the Multi-Agent Engine. It shows how smaller components work together to complete tasks. The Agent Manager manages agents, the Communication Module shares information, the State Analyzer examines game conditions, and the Strategy Decision Module selects actions. The Outcome Evaluator checks the results, while the Learning Module helps improve future decisions. This level explains the responsibilities and interactions of individual components

C4 LEVEL 4:



This diagram presents the internal modules of the Learning Module, including PolicyNetwork, ExperienceBuffer, RewardCalculator and PolicyUpdater. These modules process game experiences, calculate rewards and update agent policies to improve decision-making during future gameplay.

The Code level explains the detailed implementation of a selected component, which is the Learning Module in this project. It shows how the learning process is organized internally. The Policy Network helps select suitable actions, the Experience Buffer stores previous experiences, the Reward Calculator calculates rewards based on results, and the Policy Updater updates strategies. These operations help agents learn from past experiences and make better decisions in future games.